Subject: Mathematics | **WEEKLY CHAPTER TEST – 04 (PROBABILITY)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 14: Probability** | **Max. Marks: 20**

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

Question 1 [1 Mark]

Two dice are rolled together. The probability of obtaining a total score of 7 is:

(a) $\frac{1}{6}$ (b) $\frac{7}{36}$

(c) $\frac{5}{36}$ (d) $\frac{1}{12}$

Working & Calculation:

- When two dice are thrown simultaneously, total elementary outcomes: $n(S) = 6 \times 6 = 36$.
- Let E be the event of obtaining a sum equal to 7:

$$E = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\} \implies n(E) = 6$$

- Therefore:

$$P(\text{total score of } 7) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{1}{6}$

Question 2 [1 Mark]

All the red face cards are removed from a well-shuffled pack of 52 cards. A card is then drawn at random from the remaining cards. The probability that the drawn card is a red card is:

(a) $\frac{20}{46}$ (b) $\frac{10}{23}$

(c) $\frac{1}{2}$ (d) $\frac{13}{23}$

Working & Calculation:

- Red face cards in a deck: 3 Hearts (J, Q, K) +3 Diamonds (J, Q, K) = 6 cards removed.
- Total remaining cards: $n(S') = 52 - 6 = 46$.
- Number of remaining red cards: $26 - 6 = 20$.
- Therefore, the required probability is:

$$P(\text{red card}) = \frac{20}{46} = \frac{10}{23}$$

- In standard fractional lowest terms, this corresponds to Option (b) $\frac{10}{23}$ [equivalent to Option (a) $\frac{20}{46}$].

CORRECT ANSWER: OPTION (b) $\longrightarrow \frac{10}{23}$ [Equivalent to (a) $\frac{20}{46}$]

Question 3 [1 Mark]

A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, the probability that it bears a two-digit number is:

(a) $\frac{9}{10}$ **(b)** $\frac{89}{90}$

(c) $\frac{81}{90}$ **(d)** $\frac{4}{5}$

Working & Calculation:

- Total discs in the box: $n(S) = 90$.
- Single-digit numbers are $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ (9 discs).
- Number of two-digit numbers (10 through 90): $90 - 9 = 81$.

$$P(\text{two-digit number}) = \frac{81}{90} = \frac{9}{10}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow$ $\frac{9}{10}$ [Equivalent to (c) $\frac{81}{90}$]

Question 4 [1 Mark]

A box contains 400 apples. The probability of getting a rotten apple in the box is 0.035. The total number of rotten apples in the box is:

(a) 7 **(b) 14**

(c) 21 **(d) 28**

Working & Calculation:

- Total number of apples: $n(S) = 400$.
- Probability of drawing a rotten apple: $P(\text{rotten}) = 0.035$.
- By the definition of empirical probability:

$$P(\text{rotten}) = \frac{\text{Number of rotten apples}}{n(S)} \implies \text{Number of rotten apples} = 0.035 \times 400 = 14$$

CORRECT ANSWER: OPTION (b) $\longrightarrow$ 14

Question 5 (Assertion–Reason) [1 Mark]

Assertion (A): The probability of drawing a face card from a well-shuffled pack of 52 cards is $\frac{3}{13}$.
Reason (R): In a deck of 52 cards, there are 12 face cards (4 Kings, 4 Queens, 4 Jacks).

Options:

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

Step-by-Step Logical Analysis:

1. **Reason (R):** A standard 52-card pack has 4 suits, each containing 3 face cards (Jack, Queen, King). Total face cards $= 4 \times 3 = 12$. $\implies$ **(R) is TRUE.**
2. **Assertion (A):** Using the 12 face cards identified in (R), $P(\text{face card}) = \frac{12}{52} = \frac{3}{13}$. $\implies$ **(A) is TRUE.**
3. **Explanation Check:** The probability in (A) is directly calculated from the card count stated in (R). Hence, (R) correctly explains (A).

CORRECT ANSWER: OPTION (a) $\longrightarrow$ Both (A) and (R) are true, and (R) is the correct explanation of (A).

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [2 Marks]

Question 6 [2 Marks]

A bag contains 5 red, 8 green, and 7 white balls. One ball is drawn at random from the bag. What is the probability that the ball drawn is: (i) white (ii) not green?

Sample Space: Total number of balls in the bag: $n(S) = 5 + 8 + 7 = 20$.

(i) Probability that the ball drawn is white [1 Mark]

Number of white balls: $n(\text{white}) = 7$:

$$P(\text{white}) = \frac{n(\text{white})}{n(S)} = \frac{7}{20}$$

Answer (i): $\mathbf{P(\text{white}) = \frac{7}{20}}$

(ii) Probability that the ball drawn is not green [1 Mark]

Favourable balls (balls that are red or white): $5 + 7 = 12$ balls [or $20 - 8 = 12$]:

$$P(\text{not green}) = \frac{12}{20} = \frac{3}{5} \quad \left[\text{or via complement: } 1 - \frac{8}{20} = 1 - \frac{2}{5} = \frac{3}{5}\right]$$

Answer (ii): $\mathbf{P(\text{not green}) = \frac{3}{5}}$

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [9 Marks]

Question 7 [3 Marks]

Three unbiased coins are tossed together. Find the probability of getting:
(i) all heads (ii) at least two heads (iii) at most two heads.

Sample Space: When three coins are tossed, $n(S) = 2^3 = 8$ outcomes:

$$S = \{\text{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}\}$$

(i) All heads [1 Mark]

Favourable outcome: $E_1 = \{\text{HHH}\} \implies n(E_1) = 1$:

$$P(\text{all heads}) = \frac{n(E_1)}{n(S)} = \frac{1}{8}$$

Answer (i): $\mathbf{P(\text{all heads}) = \frac{1}{8}}$

(ii) At least two heads [1 Mark]

"At least two heads" means 2 heads or 3 heads: $E_2 = \{\text{HHT, HTH, THH, HHH}\} \implies n(E_2) = 4$:

$$P(\text{at least two heads}) = \frac{n(E_2)}{n(S)} = \frac{4}{8} = \frac{1}{2}$$

Answer (ii): $\mathbf{P(\text{at least two heads}) = \frac{1}{2}}$

(iii) At most two heads [1 Mark]

"At most two heads" means 0, 1, or 2 heads (all outcomes except HHH): $n(E_3) = 8 - 1 = 7$:

$$P(\text{at most two heads}) = \frac{7}{8} \quad \left[\text{or via complement: } 1 - P(\text{all heads}) = 1 - \frac{1}{8} = \frac{7}{8}\right]$$

Answer (iii): $\mathbf{P(\text{at most two heads}) = \frac{7}{8}}$

Question 8 [3 Marks]

The king, queen, and jack of clubs are removed from a deck of 52 playing cards and the remaining cards are well shuffled. A card is drawn at random from the remaining cards. Find the probability of getting:
(i) a club (ii) a face card (iii) a red card.

Given Setup: 3 cards (King, Queen, Jack of clubs) removed $\implies$ Remaining cards $n(S') = 52 - 3 = 49$.

(i) A club [1 Mark]

Originally there were 13 clubs. With 3 club cards removed, remaining clubs $= 13 - 3 = 10$:

$$P(\text{a club}) = \frac{10}{n(S')} = \frac{10}{49}$$

Answer (i): $\mathbf{P(\text{a club}) = \frac{10}{49}}$

(ii) A face card [1 Mark]

Originally there were 12 face cards. Since 3 face cards (K, Q, J of clubs) were removed, remaining face cards $= 12 - 3 = 9$:

$$P(\text{a face card}) = \frac{9}{n(S')} = \frac{9}{49}$$

Answer (ii): $\mathbf{P(\text{a face card}) = \frac{9}{49}}$

(iii) A red card [1 Mark]

All 3 removed cards belonged to the black suit (Clubs). No red cards were removed $\implies$ Red cards remaining $= 26$:

$$P(\text{a red card}) = \frac{26}{n(S')} = \frac{26}{49}$$

Answer (iii): $\mathbf{P(\text{a red card}) = \frac{26}{49}}$

Question 9 [3 Marks]

A box contains 90 discs numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears:
(i) a two-digit number (ii) a perfect square number (iii) a number divisible by 5.

Sample Space: Discs are numbered $1, 2, \ldots, 90 \implies n(S) = 90$.

(i) A two-digit number [1 Mark]

Two-digit numbers are $10, 11, \ldots, 90 \implies n(E_1) = 90 - 9 = 81$:

$$P(\text{two-digit number}) = \frac{81}{90} = \frac{9}{10}$$

Answer (i): $\mathbf{P(\text{two-digit number}) = \frac{9}{10}}$

(ii) A perfect square number [1 Mark]

Perfect squares from 1 to 90 are $E_2 = \{1, 4, 9, 16, 25, 36, 49, 64, 81\} \implies n(E_2) = 9$:

$$P(\text{perfect square}) = \frac{9}{90} = \frac{1}{10}$$

Answer (ii): $\mathbf{P(\text{perfect square}) = \frac{1}{10}}$

(iii) A number divisible by 5 [1 Mark]

Multiples of 5 in $\{1, \ldots, 90\}$ are $E_3 = \{5, 10, 15, \ldots, 90\} \implies n(E_3) = \frac{90}{5} = 18$:

$$P(\text{divisible by 5}) = \frac{18}{90} = \frac{1}{5}$$

Answer (iii): $\mathbf{P(\text{divisible by 5}) = \frac{1}{5}}$

SECTION D: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 10 (Case Study Analysis) [4 Marks]

"Eight Ball" is a popular cue sport played on a pool table with 15 object balls numbered 1 to 15, and one white cue ball. The object balls are partitioned as follows:
· Balls 1 to 7 are solid colored.
· Ball 8 is solid black.
· Balls 9 to 15 are striped.
On a break shot, one object ball is pocketed at random. Based on the above scenario, answer the following:



Classification of Object Balls: Total object balls $n(S) = 15$.

- Solid balls: $\{1, 2, 3, 4, 5, 6, 7\}$ (7 balls).
- Eight ball: $\{8\}$ (1 ball).
- Striped balls: $\{9, 10, 11, 12, 13, 14, 15\}$ (7 balls).

(i) Probability that the pocketed ball is a solid colored ball (from 1 to 7) [1 Mark]

Favourable balls = 7 (balls $1, 2, 3, 4, 5, 6, 7$):

$$P(\text{solid ball}) = \frac{7}{15}$$

Answer (i): $\mathbf{P(\text{solid colored ball}) = \frac{7}{15}}$

(ii) Probability that the pocketed ball is an even-numbered striped ball [1 Mark]

Striped balls are $\{9, 10, 11, 12, 13, 14, 15\}$. The even numbers in this subset are $\{10, 12, 14\}$ (3 balls):

$$P(\text{even-numbered striped}) = \frac{3}{15} = \frac{1}{5}$$

Answer (ii): $\mathbf{P(\text{even striped ball}) = \frac{1}{5}}$

(iii) Probability that the pocketed ball has a prime number [1 Mark]

Prime numbers among $\{1, 2, 3, \ldots, 15\}$ are $\{2, 3, 5, 7, 11, 13\}$ (6 prime balls):

$$P(\text{prime-numbered ball}) = \frac{6}{15} = \frac{2}{5}$$

Answer (iii): $\mathbf{P(\text{prime number}) = \frac{2}{5}}$

(iv) Probability that the pocketed ball is the 8-ball [1 Mark]

There is exactly 1 ball bearing the number 8:

$$P(\text{the 8-ball}) = \frac{1}{15}$$

Answer (iv): $\mathbf{P(\text{the 8-ball}) = \frac{1}{15}}$

— END OF WORKED SOLUTIONS (TEST–04) · MATHS MANTHAN TUITION CLASSES —